

Paper I - Basic Sciences in Urology (MCh Urology, Summer 2015)

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I-1. Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks) | ~1500 words

Introduction

- Urine formation = **glomerular filtration** → **tubular reabsorption** → **tubular secretion** → **concentration/dilution**; transport = active **peristalsis** of pelvis/ureter to a low-pressure, compliant bladder.
- Sources: Campbell: Renal Physiology and Pathophysiology chapters; Physiology and Pharmacology of the Renal Pelvis and Ureter; Pathophysiology of Urinary Tract Obstruction.

A. Renal blood flow (RBF)

- RBF ≈ **20-25% of cardiac output** (~1-1.2 L/min); RPF ~600 mL/min; cortex receives ~90%, medulla <10% (protects medullary gradient but makes it hypoxia-prone).
- Serial arterioles: **afferent** (regulates inflow) → glomerulus → **efferent** (maintains glomerular pressure) → peritubular capillaries / **vasa recta**.
- **Autoregulation** keeps RBF and GFR constant over MAP ~**80-180 mmHg**:
 - **Myogenic** - afferent smooth muscle contracts on stretch.
 - **Tubuloglomerular feedback (TGF)** - ↑ NaCl at **macula densa** → adenosine/ATP → afferent constriction (and ↓ renin).
- Extrinsic: sympathetic (α1 vasoconstriction, β1 renin), Ang II (efferent > afferent), endothelin, thromboxane (constrictors); PGE2/PGI2, NO, bradykinin, ANP (dilators). NSAIDs remove PG protection in low-flow states.

B. Glomerular filtration

- GFR ~**125 mL/min (~180 L/day)**; filtration fraction (GFR/RPF) ≈ **20%**.
- **Starling equation:** $GFR = K_f \times [(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$.
- Determinants: glomerular capillary hydrostatic pressure (main), Bowman's space pressure (↑ in **obstruction**), plasma oncotic pressure, Kf (surface area - mesangial contraction; permeability).
- Filtration barrier: fenestrated endothelium → GBM (negative charge, heparan sulphate) → podocyte slit diaphragm (nephrin, podocin). Restricts by size (>~ 60-70 kDa) and charge (albumin excluded).

- Measurement: inulin (gold standard), creatinine clearance, eGFR (CKD-EPI), DTPA renography (split function).
- Tubular secretion (PCT organic anion/cation transporters) eliminates drugs, contrast and creatinine – basis of MAG3 renography (secreted) vs DTPA (filtered).

C. Tubular handling – segment-wise

Segment	Key transporters	Handling	Clinical/drug link
PCT	Na ⁺ /H ⁺ (NHE3), SGLT2/1, Na ⁺ /K ⁺ -ATPase, AQP1, carbonic anhydrase	~ 65% Na ⁺ /water (iso-osmotic), ~100% glucose & amino acids, ~80–90% HCO ₃ ⁻ , most phosphate (PTH ↓), secretes organic acids/bases (drugs, creatinine)	Acetazolamide, SGLT2i; Fanconi syndrome; cystinuria (dibasic AA)
Thin descending limb	AQP1	Water-permeable, solute-impermeable → tubular fluid concentrates	
Thin ascending limb	–	Water-impermeable; passive NaCl exit	
Thick ascending limb (TAL)	NKCC2 , ROMK, Ca/Mg paracellular (claudin-16/19)	~ 25% NaCl; impermeable to water → " diluting segment "; generates medullary gradient	Loop diuretics; Bartter; secretes Tamm–Horsfall (uromodulin)
DCT	NCC , TRPV5 (Ca ²⁺)	~5–7% Na ⁺ ; active Ca ²⁺ reabsorption (PTH)	Thiazides ↓ urinary Ca (hypercalciuric stones); Gitelman
Collecting duct – principal cells	ENaC , ROMK, AQP2 (apical), AQP3/4 (basolateral)	Na ⁺ reabsorption/K ⁺ secretion (aldosterone); water (ADH)	Amiloride, spironolactone; Liddle
Intercalated cells	Type A: H ⁺ -ATPase; Type B: pendrin	Acid/base	Distal RTA (type 1) → CaP stones, nephrocalcinosis

D. Concentration and dilution

- **Countercurrent multiplication (loop of Henle):** TAL pumps NaCl out without water → single effect (~200 mOsm gradient at each level) multiplied by countercurrent flow → medullary interstitium up to ~**1200 mOsm/kg** at papilla.
- **Urea recycling:** ADH-stimulated urea transporters (UT-A1/A3) in inner medullary collecting duct contribute ~half of inner-medullary osmolality.
- **Countercurrent exchange (vasa recta):** hairpin capillaries, slow flow – preserve the gradient (washout in osmotic diuresis/high flow).
- **ADH (vasopressin):** released from posterior pituitary for ↑ plasma osmolality (~1–2% change sensed) or ↓ volume → **V2 receptor** on basolateral principal cell → Gs → cAMP → PKA → **phosphorylation and insertion of AQP2** vesicles into apical membrane → water reabsorption along gradient. Long-term ↑ AQP2 synthesis.
 - V1a – vasoconstriction; V2 also on endothelium (vWF/factor VIII release – basis of **desmopressin** in nocturia/enuresis/bleeding).
 - Urine osmolality range ~**50–1200 mOsm/kg**.
 - Nephrogenic DI: V2/AQP2 mutations, lithium, hypercalcaemia, hypokalaemia, **post-obstructive** (↓ AQP2).

E. Hormonal control (summary)

Hormone	Site	Effect
Ang II / aldosterone	PCT, efferent; CD (ENaC)	Na ⁺ retention, K ⁺ /H ⁺ loss, maintains GFR
ADH	CD	Free-water reabsorption
ANP/BNP	Afferent dilatation, CD	Natriuresis, ↑ GFR, ↓ renin/aldosterone
PTH	PCT (↓ PO ₄), DCT (↑ Ca)	Phosphaturia, Ca retention; 1α-hydroxylase
Prostaglandins, NO, dopamine	Vessels, tubule	Vasodilatation, natriuresis

F. Upper urinary tract transport

1. Pacemaker activity

- Site: **proximal renal pelvis/pelviccalyceal border** (minor calyces in multicalyceal kidneys).
- **Atypical smooth muscle cells** - primary pacemakers (spontaneous depolarisation; HCN/T-type Ca²⁺ currents implicated).
- **Interstitial cells of Cajal-like cells (c-Kit⁺)** - in pelvis and ureter; thought to act as secondary pacemakers/conductors coupling atypical SMCs to typical SMCs; may drive activity distally if proximal pacemaker lost (relevance to **UPJ obstruction** - reduced ICC-like cells reported).
- Activity is **myogenic**: peristalsis persists after denervation and in transplanted kidneys; autonomic nerves only modulate.

2. Electromechanical coupling

- Typical ureteric SMC resting potential approx. -33 to -70 mV.
- Action potential: rapid upstroke (**L-type Ca²⁺ influx**) → **plateau** with oscillations → repolarisation (K⁺ efflux).
- Spread cell-to-cell via **gap junctions** (functional syncytium) at ~2-6 cm/s.
- Ca²⁺ + calmodulin → **myosin light-chain kinase** → phosphorylation → actin-myosin cross-bridging → contraction. Relaxation via cAMP/cGMP (β-agonists, PDE inhibitors, NO).

3. Bolus propagation and pressures

- Pelvis fills → pelvic pressure rises → urine pushed through UPJ → ureter contracts proximally behind a **bolus**; walls **coapt** fully so urine moves as discrete boluses.
- Peristalsis ~**2-6/min**; resting ureteral pressure ~0-5 cmH₂O; contraction waves ~20-80 cmH₂O.
- At low flow, pelvis contracts more often than ureter ("**pyeloureteric block**" - not every pelvic contraction propagates).

4. Effect of flow rate

- ↑ Flow → ↑ frequency first, then ↑ bolus volume; at very high flow boluses **coalesce** → walls no longer coapt → continuous **fluid column**, ureter dilates - transport becomes inefficient.
- Hence dilatation can occur **without obstruction** (diuresis, DI, high-flow states - relevant to diuretic renogram interpretation).

5. Ureterovesical junction (UVJ)

- Oblique intramural tunnel, Waldeyer sheath; **tunnel length:diameter ≈ 5:1** (Paquin) prevents reflux.
- Ureteric bolus pressure must exceed intravesical pressure; with sustained high bladder pressure (>~40 cmH₂O - neurogenic bladder) efflux fails → **secondary hydronephrosis**/upper tract damage.

6. Factors affecting transport

Factor	Effect
Obstruction	Acute: ↑ pressure & ↑ contraction amplitude; chronic: dilatation, ↓ coaptation, decompensation, fibrosis
Infection	Bacterial endotoxin (Gram-negative) inhibits peristalsis → dilatation without obstruction
Pregnancy	Dilatation (R > L) mainly mechanical compression at pelvic brim; progesterone contributes
Drugs	α-agonists stimulate; α1-blockers (tamsulosin – MET), β-agonists, Ca-channel blockers (nifedipine), PDE5i relax; NSAIDs ↓ PG-mediated pelvic pressure/diuresis (renal colic); opioids variable
Age/VUR	Fetal/infant ureter compliant; reflux → dilatation

G. Bladder storage (brief)

- Accommodation with low pressure: **high compliance** (viscoelastic wall, collagen/elastin) + **sympathetic** β3 relaxation of detrusor, α1 contraction of bladder neck (T10–L2, hypogastric); **pudendal** (Onuf's nucleus, S2–4) external sphincter tone (guarding reflex); pontine micturition centre inhibited.
- Normal capacity ~400–500 mL; storage pressure normally low (<~15 cmH₂O at capacity).

H. Pathophysiology of obstruction

Unilateral ureteral obstruction (UUO) – triphasic response (Campbell; Vaughan/Moe canine studies):

Phase 1 (0 – ~1.5 h): RBF ↑, ureteral pressure ↑
 - afferent vasodilatation (PGE₂, NO)
 Phase 2 (~1.5 – 5 h): RBF ↓, ureteral pressure still ↑ (keeps rising)
 - efferent vasoconstriction begins
 Phase 3 (> 5 h, persists): RBF ↓, ureteral pressure ↓
 - afferent (preglomerular) vasoconstriction: Ang II, thromboxane A₂, endothelin

- GFR falls because of ↓ RBF and ↑ tubular pressure; contralateral kidney shows compensatory ↑ function.
- Bilateral obstruction/solitary kidney:** RBF rises only modestly then falls; **ureteral pressure stays high longer**; accumulation of natriuretic substances (ANP, urea) → **post-obstructive diuresis** after relief.
- Tubular effects:** ↓ concentrating ability (↓ AQP2, washout), ↓ Na reabsorption, **distal RTA**, ↓ K⁺ secretion.
- Chronic:** Ang II and TGF-β1 → macrophage infiltration, tubular apoptosis, interstitial fibrosis, **tubular atrophy**. Recovery depends on duration, degree, compliance of pelvis (intrarenal pelvis worse), infection.
- Pyelorenal backflow** (pyelotubular, pyelovenous, pyelolymphatic, pyelointerstitial) acts as pop-off and allows partial function under obstruction; forniceal rupture → urinoma.

Diagram to draw: (1) Nephron with segment-wise transporters and % Na⁺ reabsorbed, with countercurrent loop and ADH/AQP2 signalling inset. (2) Triphasic graph of RBF and ureteral pressure vs time in UUO.

Recent advances

- SGLT2 inhibitors – restore TGF (afferent constriction) → nephroprotection (also increase glycosuria; UTI/genital infection signal).
- Vaptans (V2 antagonists: tolvaptan) – aquaresis; used in ADPKD.
- Biomarkers of obstructive injury: urinary NGAL, KIM-1, MCP-1 (research/paediatric UPJO follow-up).
- ICC-like cell and HCN channel research in pathogenesis of congenital UPJO.

Key points to remember: - RBF ~20–25% CO; GFR 125 mL/min; FF 20%; autoregulation 80–180 mmHg (myogenic + TGF). - PCT 65%, TAL 25% (water-impermeable diluting segment), DCT 5–7%, CD fine-tunes via aldosterone/ADH (V2 → cAMP → AQP2). - Pacemaker = atypical SMCs at pelvicalyceal border; ICC-like cells modulate; peristalsis is myogenic, 2–6/min. - High flow → bolus coalescence → dilatation without obstruction. - UUO triphasic: ↑/↑, ↓/↑, ↓/↓ (RBF/pressure).

I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks) | ~1500 words

Introduction

- Pelvic organ support depends on an **interaction of muscles (levator ani), connective tissue (endopelvic fascia and ligaments) and bony pelvis**. Failure → **POP** (descent of anterior/apical/posterior vaginal wall) and **SUI** (involuntary leakage on effort when urethral support/closure fails).
- Campbell: Anatomy of the lower urinary tract/female pelvic floor; Pelvic organ prolapse; Stress incontinence chapters.

A. Bony pelvis

- Female pelvis wider, shallower, gynaecoid inlet; in standing posture the pubic rami and ischial spines lie almost in the same horizontal plane → pelvic organs rest on the **levator plate** rather than the hiatus.
- Key landmarks for surgery: **ischial spine** (ATFP, ATLA and sacrospinous ligament attach), **sacrospinous ligament**, **Cooper's (iliopectineal) ligament**, **sacral promontory/anterior longitudinal ligament**, obturator foramen.

B. Pelvic floor muscles (pelvic diaphragm)

Component	Origin → insertion	Function
Pubococcygeus/pubovisceral (pubovaginalis, puboperinealis, puboanalis)	Pubis → vagina, perineal body, anus	Closes hiatus, lifts urethra/vagina forward
Puborectalis	Pubis → sling behind anorectal junction	Anorectal angle; closes urogenital hiatus
Iliococcygeus	Arcus tendineus levator ani (ATLA) on obturator fascia → anococcygeal raphe/coccyx	Forms horizontal levator plate
(Coccygeus/ischiococcygeus)	Ischial spine → sacrum/coccyx	Posterior support; overlies sacrospinous ligament

- Constant **basal tone** (type I fibres) keeps the **levator hiatus** closed; reflex contraction on cough.
- **Levator avulsion** (pubovisceral detachment at vaginal delivery, best seen on 3D/4D translabial USG or MRI) and ↑hiatal area are strong risk factors for POP and recurrence.
- **Perineal membrane (urogenital diaphragm)** and perineal body – distal support (level III).

C. Endopelvic fascia and ligaments

- Fibromuscular connective tissue (collagen, elastin, smooth muscle) enveloping viscera:
 - Cardinal (Mackenrodt's) & uterosacral ligaments** – apex suspension.
 - Pubocervical fascia** – anterior vaginal wall "fascia" attached laterally to **arcus tendineus fasciae pelvis (ATFP, white line)** (pubis → ischial spine).
 - Rectovaginal fascia (Denonvilliers)** – posterior, attached to perineal body below and uterosacral complex above.
 - Pubourethral ligaments** – mid-urethra to pubis (Petros; Zacharin).
- Histologically the vaginal "fascia" is the fibromuscular layer of the vaginal wall – no true separate fascia (important when describing "fascial" repairs).

D. DeLancey's three levels of vaginal support (DeLancey 1992)

Level	Region	Structures	Defect →
I - suspension	Upper vagina/cervix	Cardinal-uterosacral complex	Uterine/vault prolapse, enterocele
II - attachment	Mid vagina	Lateral attachment to ATFP (anterior) and ATLA/rectovaginal fascia (posterior)	Cystocele (paravaginal), rectocele
III - fusion	Distal ~2-3 cm	Fusion to perineal membrane, perineal body, levator	Urethral hypermobility, perineal descent, low rectocele

E. Urethral support and continence mechanisms

1. Hammock hypothesis (DeLancey 1994)

- Urethra lies on a supportive layer of **anterior vaginal wall + endopelvic fascia**, attached laterally to ATFP and levator ani.
- Rise in abdominal pressure **compresses urethra against this firm hammock** → closure. Loss of the hammock (laxity/detachment) → urethra not compressed → SUI. (Replaces Enhörning's older "pressure transmission" theory.)

2. Integral theory (Petros & Ulmsten, 1990)

- Stress and urge symptoms both arise from **laxity of the vagina or its supporting ligaments** (pubourethral, uterosacral) due to altered connective tissue.
- Three directional muscle forces (anterior pubococcygeus, levator plate, longitudinal muscle of anus) act around ligaments; **mid-urethral pubourethral ligament** is the key – basis of the **tension-free vaginal tape (TVT, Ulmsten 1996)** placed at mid-urethra.

3. Intrinsic urethral closure (sphincter)

- Mucosal coaptation (oestrogen-dependent vascular submucosal plexus), smooth muscle, and striated **rhabdosphincter**: sphincter urethrae (middle third) + **compressor urethrae and urethrovaginal sphincter** (distal arching fibres).
- Maximal urethral closure pressure falls with age; **intrinsic sphincter deficiency (ISD)** → low VLPP (<60 cmH₂O)/MUCP (<20 cmH₂O) – poorer results with colposuspension; prefer retropubic sling/autologous sling/bulking.
- SUI = **hypermobility** (support failure) ± **ISD** – a continuum, not separate entities.

F. Compartments and defects

Compartment	Defect	Anatomical basis
Anterior	Central cystocele (distension)	Midline attenuation of pubocervical fascia – rugae lost
	Lateral (paravaginal) cystocele (displacement)	Detachment from ATFP – rugae preserved, lateral sulci lost
	Transverse/apical	Separation from pericervical ring
Apical	Uterine/vault prolapse, enterocele	Level I – uterosacral/cardinal failure; enterocele = peritoneum herniation between vaginal fascia layers (post-hysterectomy, after colposuspension)
Posterior	Rectocele, perineal descent	Rectovaginal fascia defect, perineal body disruption

- Coexistence common: **apical support underpins anterior** – anterior repair fails if apex not addressed.

G. Innervation

- **Pudendal nerve (S2-4)** – external urethral/anal sphincters, perineal muscles; traverses Alcock's canal; stretch/compression injury in delivery.
- **Levator ani nerve (S3-5)** – runs on the **pelvic (superior) surface** of levator, separate from pudendal (DeLancey/Barber).
- Autonomic: pelvic splanchnic (S2-4, parasympathetic), hypogastric (T10-L2) – injured in radical pelvic surgery.
- Neuropathy (denervation of pelvic floor) after childbirth contributes to POP/SUI.

H. Pathophysiology – risk factors (Bump & Norton)

- **Predisposing:** genetics, race, connective tissue disorders (Ehlers-Danlos, Marfan), collagen type III > I.
- **Inciting: vaginal childbirth** (strongest; forceps, macrosomia, prolonged 2nd stage, levator avulsion), hysterectomy, pelvic surgery.
- **Promoting:** obesity, chronic cough/COPD, constipation, heavy lifting, smoking.
- **Decompensating:** ageing, **menopause (oestrogen loss)**, neuropathy, myopathy.

I. POP-Q staging (ICS, Bump et al 1996)

- Reference = **hymen (0)**; above = negative, below = positive (cm). Nine measurements:

Point	Description	Range
Aa	Anterior wall 3 cm proximal to hymen	-3 to +3
Ba	Most dependent anterior wall (Aa-C)	-3 to +tvI
C	Cervix/vault	
D	Posterior fornix (omitted post-hysterectomy)	
Ap / Bp	Posterior equivalents of Aa/Ba	
gh	Genital hiatus (urethral meatus → posterior hymen)	
pb	Perineal body	
tvI	Total vaginal length	

Stage	Definition
0	No prolapse (Aa, Ap, Ba, Bp = -3; C/D ≤ -(tvI-2))
I	Most distal point > 1 cm above hymen (< -1)
II	Most distal point within ±1 cm of hymen
III	> +1 cm below hymen but < (tvI - 2) cm
IV	Complete eversion (≥ tvI - 2)

Clinical assessment linked to anatomy

- Examination supine and standing with Valsalva; Sims speculum to isolate each compartment; **cough stress test** for SUI (also after reducing prolapse - **occult SUI**).
- **Q-tip test**: straining angle change >30° = urethral hypermobility.
- Lateral sulci preserved vs lost (paravaginal vs central defect); rectovaginal examination for enterocele/rectocele; levator strength (Oxford scale) and palpable avulsion.
- Urodynamics (VLPP/MUCP) when ISD or mixed symptoms suspected; translabial USG/dynamic MRI in complex/recurrent cases.

J. How anatomy guides surgery

Defect/aim	Operation	Anatomical principle
SUI - hypermobility	Mid-urethral sling (retropubic TVT / transobturator TOT)	Integral theory - recreates pubourethral ligament support at mid-urethra
SUI	Burch colposuspension	Paravaginal/periurethral fascia at bladder neck level → Cooper's ligament (restores hammock)
SUI - ISD	Autologous fascial (pubovaginal) sling at bladder neck, bulking, AUS	Compression/coaptation
Paravaginal cystocele	Paravaginal repair	Reattach to ATRP (level II)
Central cystocele	Anterior colporrhaphy	Midline plication
Apical (level I)	Sacrocolpopexy (open/lap/robotic - gold standard durability)	Mesh vault/vagina → anterior longitudinal ligament over S1 promontory (beware middle sacral vessels, left common iliac vein)
Apical vaginal route	Sacrospinous ligament fixation (usually right; suture ~1.5-2 cm medial to ischial spine to avoid pudendal vessels/nerve), uterosacral ligament suspension (ureteric injury risk - check cystoscopy)	Level I restoration
Rectocele	Posterior colporrhaphy/site-specific repair, perineorrhaphy	Level II/III
Elderly, not sexually active	Colpocleisis	Obliteration

K. Mesh regulation

- **FDA, April 2019**: ordered manufacturers (Boston Scientific, Coloplast - last remaining) to **stop selling and distributing transvaginal mesh for anterior-compartment POP repair** (lack of proven benefit-risk). Did **not** apply to mid-urethral slings or sacrocolpopexy mesh.
- UK (2018) paused vaginal mesh including MUS pending safeguards (Cumberlege review 2020). Mesh complications: exposure/erosion, pain, dyspareunia - counsel and document.

Diagram to draw: (1) DeLancey levels I-III on a lateral/3D vaginal sketch with ATFP and uterosacral ligaments. (2) Hammock: cross-section of urethra on anterior vaginal wall attached to ATFP, with abdominal pressure arrow.

Recent advances

- 3D/4D translabial ultrasound and dynamic MRI – levator avulsion, hiatal ballooning (predict recurrence).
- Robotic sacrocolpopexy and hysteropexy (uterine-sparing) increasing.
- Single-incision mini-slings – comparable short-term outcomes to standard MUS (UK SIMS RCT, 2022).
- Autologous fascial sling and colposuspension resurgence after mesh concerns; laser therapies for SUI remain experimental (FDA warning 2018 on energy-based vaginal devices).

Key points to remember: - DeLancey levels: I suspension (cardinal-uterosacral), II attachment (ATFP), III fusion (perineal membrane/body). - Hammock (DeLancey 1994) vs Integral theory (Petros & Ulmsten 1990 → TVT 1996, mid-urethra). - Central (rugae lost) vs paravaginal (rugae preserved) cystocele. - POP-Q: hymen = 0; Stage II = ± 1 cm of hymen; Stage IV $\geq$ tvl-2. - FDA 2019 – transvaginal mesh for anterior POP withdrawn; MUS and sacrocolpopexy mesh unaffected.

I-3a. Newer generation CT scan

Type: SAQ (10 marks) | ~550 words

Introduction

- CT is the reference imaging for stones, renal masses, haematuria (CTU) and trauma.
Campbell: Urinary tract imaging chapter; EAU Urolithiasis and Upper tract UC guidelines.

Evolution

Generation	Feature
Conventional (Hounsfield, 1971–72)	Step-and-shoot axial
Helical/spiral CT (~1989)	Continuous table movement + slip-ring gantry → volumetric data, no misregistration
MDCT (4-slice 1998 → 16/64 → 256/320-row)	Sub-millimetre isotropic voxels, multiplanar/3D reconstruction, single breath-hold
Dual-source CT (2005)	Two tube-detector pairs: speed, dual-energy
Photon-counting CT (PCCT)	First FDA clearance 2021 (Siemens NAEOTOM Alpha)

Dual-energy CT (DECT)

- Acquires at two energy spectra (e.g. 80/140 kVp) via dual-source, rapid kV switching, dual-layer detector or split filter.
- Material decomposition based on attenuation ratio:
 - **Uric acid vs non-uric-acid stones** (accuracy high) → selects **oral chemolysis** (alkalinisation) candidates; also cystine, struvite partial differentiation.
 - **Virtual non-contrast (VNC)** images → drop the true unenhanced phase (dose reduction).
 - **Iodine maps** – enhancement of small/hyperdense cysts vs solid masses; stones visible within contrast-filled collecting system.

- Reduced metal/beam-hardening artefact (virtual monoenergetic images).

Low/ultra-low dose CT KUB

- Standard NCCT abdomen-pelvis ~**8-10 mSv**; **low-dose** CT KUB **<3-4 mSv**; **ultra-low-dose** **<1 mSv** (comparable to plain KUB+).
- Sensitivity/specificity for stones remain ~95% (lower for small stones <3 mm and BMI >30). EAU recommends **low-dose NCCT** for acute flank pain (after USG) and follow-up; avoid repeated standard-dose CT in young/recurrent stone formers.

CT urography (CTU)

- Indicated for **visible haematuria/UTUC** evaluation (EAU).
- **Triple-phase**: unenhanced → nephrographic (~90-100 s) → excretory (~10-15 min) – dose ~15-25 mSv or more.
- **Split-bolus**: contrast in 2 aliquots; single post-contrast acquisition gives combined nephrographic + excretory phase → lower dose (with DECT VNC, a single acquisition possible).
- Adjuncts: IV furosemide/saline for distension; prone imaging.

Dose reduction technology

- Automatic tube current modulation, low kVp.
- **Iterative reconstruction** (statistical/model-based) – dose reduction while maintaining image quality.
- **Deep-learning reconstruction** (AI-based denoising, commercially available) – further noise reduction at low dose.

Photon-counting CT

- Detector (CdTe) counts individual photons and sorts energy → inherent spectral data, **no electronic noise**, higher spatial resolution, lower dose, better stone composition and small-structure imaging.

Other advances

- **4D/perfusion CT** – time-resolved enhancement; renal tumour characterisation, parathyroid (4D-CT), renal perfusion.
- **CT angiography** – donor nephrectomy planning, renal artery anatomy, AVM; 3D models for partial nephrectomy/PCNL planning (3D printing, virtual reality).
- **Radiomics/AI** – lesion segmentation, stone burden volumetry, prediction of SFR.
- Cone-beam CT – intraoperative in hybrid theatres for PCNL access.

Uses in urology

- Stones (size, HU >1000 suggests hard calcium oxalate monohydrate/brushite; skin-to-stone distance >10 cm → SWL failure), renal masses (Bosniak 2019 – CT/MRI based), staging (RCC, UTUC, bladder, testis), trauma (AAST grading with delayed phase), abscess/emphysematous pyelonephritis, PUJ crossing vessels.

Diagram to draw: Timeline axial → helical → MDCT → dual-source → photon-counting; CTU split-bolus timing diagram.

Recent advances

- PCCT in clinical use (since 2021); deep-learning reconstruction; AI-based stone composition prediction from single-energy CT (research).

Key points to remember: - Helical CT ~1989; MDCT 1998; dual-source 2005; PCCT FDA-cleared 2021. - DECT: uric acid vs non-UA, VNC, iodine maps. - Low-dose KUB <3-4 mSv,

ultra-low <1 mSv - EAU preferred NCCT. - Split-bolus CTU reduces phases and dose.

I-3b. History of Laparoscopy

Type: SAQ (10 marks) | ~550 words

Introduction

- Laparoscopy = endoscopic visualisation of the peritoneal cavity through a pneumoperitoneum; evolved from lighting/optics (endoscopy) → diagnostic → therapeutic → robotic. Campbell: Basics of laparoscopic and robotic urologic surgery.

Timeline

Year	Pioneer	Milestone
1806	Philipp Bozzini (Frankfurt)	"Lichtleiter" – candle-lit light conductor; birth of endoscopy
1853	Antonin Desormeaux	"Father of endoscopy"; first practical endoscope (urethra/bladder)
1877–79	Maximilian Nitze (with Leiter)	Cystoscope with distal illumination and lens system
1901	Georg Kelling (Dresden)	"Koelioskopie" – cystoscope into air-insufflated abdomen of a dog
1910	Hans Christian Jacobaeus (Stockholm)	First human laparoscopy/thoracoscopy series; coined "laparoscopy"
1911	Bernheim (Johns Hopkins)	First in USA ("organoscopy")
1924	Zollikofer	CO ₂ for insufflation
1938	János Veress (Hungary)	Spring-loaded needle (designed for therapeutic pneumothorax)
1950s	Harold Hopkins	Rod-lens system (with Karl Storz – cold light fibre-optic) → bright, clear images
1970s–80s	Kurt Semm (Kiel)	Automatic CO ₂ insufflator, endoloops, pelvi-trainer; first lap appendectomy (1980)
1970s	Harrieth Hasson	Open (Hasson) access technique
1976	Cortesi	Laparoscopy for localisation of impalpable testis – first urological use
1985	Erich Mühe (Germany)	First laparoscopic cholecystectomy
1987	Philippe Mouret (Lyon)	First video-laparoscopic cholecystectomy – popularised laparoscopy
1980s	Video chip camera	Allowed the whole team to see the image, two-handed surgery
1990	Ralph Clayman (Washington University, St Louis)	First laparoscopic nephrectomy (published 1991)
1991	Schuessler, Kavoussi et al	Laparoscopic pelvic lymphadenectomy for prostate cancer staging
1992	Gagner	Laparoscopic adrenalectomy
1992	Schuessler	First lap radical prostatectomy – reported 1997 (9 patients; "no advantage" conclusion)
1993	Schuessler; Kavoussi & Peters	Laparoscopic dismembered pyeloplasty
1995	Ratner (with Kavoussi, Johns Hopkins)	Laparoscopic live-donor nephrectomy
1998–99	Guillonneau & Vallancien (Montsouris)	Standardised LRP – revived the procedure
2000	da Vinci (Intuitive)	FDA clearance for general laparoscopic surgery (prostatectomy indication 2001)
2000	Binder & Kramer (Frankfurt); Abbou (Créteil)	First robot-assisted radical prostatectomy
Early 2000s	Mani Menon (Vattikuti Urology Institute, Detroit)	VIP – Vattikuti Institute Prostatectomy ; standardised RARP; later Retzius-sparing technique by Bocciardi (Milan)
2004	Kalloo	Transgastric peritoneoscopy (animal) → NOTES concept
2007	Raman, Rane et al	LESS/single-port nephrectomy

Year	Pioneer	Milestone
2018	da Vinci SP	FDA clearance for urological procedures (single-port robot)

Phases of evolution

Optics/illumination → Diagnostic laparoscopy → Therapeutic (Semm)
 → Video laparoscopy (lap chole boom) → Urological ablative/reconstructive
 → Robotic (3D, wristed instruments) → Single-site / NOTES / telesurgery

Indian contribution (brief)

- Rapid uptake of urological laparoscopy and robotics; indigenous robotic system (**SSI Mantra**) developed in India; telesurgery demonstrations.

Recent advances

- Multi-port competitors: Versius (CMR), Hugo RAS (Medtronic), Hinotori (Japan); ultra-long-distance telesurgery; AI-video analytics; augmented reality/3D-model overlay; haptic feedback (da Vinci 5).

Diagram to draw: Simple timeline arrow 1806 → 1901 → 1910 → 1938 → 1987 → 1990 → 2000.

Key points to remember: - Bozzini 1806 Lichtleiter; Kelling 1901 (dog); Jacobaeus 1910 (human). - Veress 1938; Hopkins rod-lens; Semm – father of modern operative laparoscopy. - Urology: Cortesi 1976 (impalpable testis); Clayman 1990 (nephrectomy); Schuessler LRP 1992/1997; pyeloplasty 1993; Ratner donor 1995. - da Vinci FDA 2000; Binder 2000 first RARP; Menon – VIP.

I-3c. Renin-angiotensin system

Type: SAQ (10 marks) | ~550 words

Definition

- Hormonal cascade regulating **blood pressure, extracellular volume, Na⁺/K⁺ balance and GFR**; systemic and local (intrarenal) components. Campbell: Renal physiology; Renovascular hypertension chapters.

Cascade

Angiotensinogen (liver, α₂-globulin)
 | Renin (JG cells) – rate-limiting step
 Angiotensin I (decapeptide, inactive)
 | ACE (lung endothelium mainly; also kidney) – also degrades bradykinin
 Angiotensin II (octapeptide) → AT₁ receptor (most actions) / AT₂ receptor
 | Aminopeptidase A | ACE2
 Angiotensin III (aldosterone release) Angiotensin (1-7) → Mas receptor
 (vasodilatation, anti-fibrotic)

Juxtaglomerular apparatus

- **JG (granular) cells** – modified smooth muscle of **afferent arteriole**; store/secrete renin; act as intrarenal baroreceptors.
- **Macula densa** – specialised TAL/early DCT cells sensing tubular NaCl.
- **Extraglomerular mesangial (Lacis) cells**.

Stimuli for renin release

Stimulate	Inhibit
↓ Renal perfusion pressure (baroreceptor)	↑ Perfusion pressure
↓ NaCl delivery to macula densa	↑ NaCl at macula densa
Sympathetic β₁ stimulation	Ang II (negative feedback)
Prostaglandins (PGE ₂ , PGI ₂)	ANP, ADH, NSAIDs, β-blockers
Loop diuretics, hypovolaemia, upright posture	High-salt diet

Receptors and actions

- **AT₁ (G_q):** arteriolar vasoconstriction; **aldosterone** release (zona glomerulosa) → ENaC Na⁺ retention, K⁺/H⁺ excretion; ADH release and thirst; ↑ PCT Na⁺/H⁺ exchange; sympathetic facilitation; cardiac/vascular hypertrophy; **fibrosis** (TGF-β, PAI-1).
- **AT₂:** vasodilatation, antiproliferative, apoptosis; important in **fetal kidney/urinary tract development** (AT₂ knockout → CAKUT incl. VUR, UPJO-like anomalies in animals).
- **ACE2/Ang(1-7)/Mas:** counter-regulatory arm; ACE2 is also the **SARS-CoV-2 entry receptor**.

Intrarenal effects on GFR

- Ang II constricts **efferent > afferent** arteriole → maintains glomerular hydrostatic pressure and GFR when perfusion is low; ↑ filtration fraction.
- Hence **ACEi/ARB → acute fall in GFR** in bilateral renal artery stenosis/stenosis in a solitary kidney, hypovolaemia, NSAID use ("triple whammy" with diuretics).
- Also contracts mesangium (↓ K_f) and enhances TGF sensitivity.

Urological relevance

- **Renovascular hypertension** (Goldblatt 1934, clip model): atherosclerotic (ostial, older men) or fibromuscular dysplasia (young women, "string of beads"). Diagnosis: Duplex, CTA/MRA, captopril renography; **renal vein renin ratio >1.5** (affected:contralateral) with suppression of contralateral predicts response to revascularisation/nephrectomy.
- **Page kidney** – perirenal compression (subcapsular haematoma post-trauma/biopsy/ESWL) → renal ischaemia → renin-mediated hypertension (Page 1939 cellophane model). Treat: drainage/decortication, ACEi, nephrectomy if refractory.
- **Juxtaglomerular cell tumour (reninoma)** (Robertson 1967): young adults, severe hypertension, **hypokalaemia, secondary hyperaldosteronism**, small cortical mass; CD34⁺/renin⁺; **partial nephrectomy curative**.
- **Obstruction:** Ang II mediates the late afferent vasoconstriction (phase 3 of UUO) and tubulo-interstitial fibrosis; ACEi/ARB attenuate fibrosis experimentally.
- Other: segmental hypoplasia (Ask-Upmark), reflux nephropathy, Wilms tumour (renin secretion), post-transplant renal artery stenosis, RCC paraneoplastic hypertension, adrenal (Conn's – aldosterone:renin ratio).
- **Drugs:** ACEi (cough, angio-oedema via bradykinin), ARBs, **direct renin inhibitor (aliskiren)**, mineralocorticoid antagonists; all contraindicated in pregnancy (fetal renal dysgenesis).

Diagram to draw: JG apparatus (afferent arteriole with JG cells, macula densa, efferent arteriole) + RAS cascade flowchart.

Recent advances

- Non-steroidal MRA **finerenone** (renal protection in diabetic CKD); aldosterone synthase inhibitors (baxdrostat) in resistant hypertension trials; angiotensinogen-silencing RNA (zilebesiran) in trials; renal denervation (FDA approval 2023) for resistant hypertension.

Key points to remember: - Renin = rate-limiting; JG cells of afferent arteriole; stimuli: ↓ pressure, ↓ macula densa NaCl, β1. - Ang II: efferent constriction maintains GFR → ACEi danger in bilateral RAS. - AT1 = pressor/fibrotic; AT2/ACE2-Ang(1-7) = protective. - Renal vein renin ratio >1.5 lateralises; reninoma → HTN + hypokalaemia, partial nephrectomy cures.

I-3d. Patho-physiology of Brain stem death

Type: SAQ (10 marks) | ~550 words

Definition

- **Irreversible loss of capacity for consciousness combined with irreversible loss of capacity to breathe**, due to permanent cessation of brainstem function – legally equivalent to death (India: THOA). The heart continues only with ventilatory support. Main source of **deceased donor (DBD)** kidneys.

Pathophysiology

Primary brain injury (TBI, SAH, ICH, hypoxic injury)
 → cerebral oedema → ↑ICP (Monro-Kellie) → ↓CPP (MAP – ICP) → ischaemia
 → rostro-caudal herniation ("coning" – uncal/tonsillar)
 → Pontine ischaemia: CUSHING REFLEX (hypertension, bradycardia, irregular breathing)
 → Medullary ischaemia: AUTONOMIC (catecholamine) STORM
 - severe hypertension, tachyarrhythmia, ↑SVR
 - myocardial contraction-band necrosis, neurogenic pulmonary oedema
 - organ ischaemia (incl. kidneys)
 → Loss of sympathetic outflow (spinal cord) → vasodilatation, HYPOTENSION, ↓cardiac output

Systemic consequences after brainstem death

System	Change	Mechanism
Cardiovascular	Hypotension, arrhythmias, LV dysfunction	Loss of vasomotor tone, catecholamine injury, hypovolaemia
Endocrine – posterior pituitary	Diabetes insipidus (majority) – polyuria, hypernatraemia	↓ADH
Anterior pituitary/thyroid/adrenal	↓T3 (sick euthyroid), relative adrenal insufficiency	Hypothalamic-pituitary failure
Pancreas	Hyperglycaemia	Insulin resistance, catecholamines, steroids
Thermoregulation	Hypothermia (poikilothermia)	Hypothalamic loss + vasodilatation
Haematology	Coagulopathy/DIC	Release of tissue thromboplastin from necrotic brain
Respiratory	Pulmonary oedema, ARDS	Neurogenic + inflammation
Immune	Systemic inflammatory response (cytokines, complement)	↑Graft immunogenicity → higher DGF and acute rejection in DBD vs living donor kidneys

Diagnosis (clinical)

1. Preconditions

- Apnoeic coma on ventilator; **known, irreversible structural cause**.

2. Exclusions

- Depressant/sedative drugs, neuromuscular blockers (check train-of-four), **hypothermia** (core temp <35 °C), severe metabolic/endocrine disturbance (Na, glucose, uraemia, hepatic), circulatory shock.

3. Brainstem reflexes (all absent)

Test	Cranial nerves
Pupils fixed, no response to light	II, III
Corneal reflex absent	V, VII
Oculovestibular (caloric - ~50 mL ice-cold water each ear, tympanic membrane visible; no eye movement)	VIII, III, VI
Oculocephalic (doll's eye) absent	VIII, III, VI
No motor response in cranial nerve distribution to central pain	V, VII
Gag and cough (tracheal suction) absent	IX, X

4. Apnoea test

- Pre-oxygenate with 100% O₂; baseline PaCO₂ normal; disconnect ventilator with O₂ insufflation via tracheal catheter; observe for respiratory effort until **PaCO₂ ≥60 mmHg** (and pH <7.30) - no breathing = positive. Abort if hypotension/desaturation/arrhythmia.
- Ancillary tests (EEG, 4-vessel angiography, transcranial Doppler, nuclear perfusion) - not mandatory in India/UK; used when clinical tests cannot be completed.

Indian legal framework

- **Transplantation of Human Organs Act (THOA), 1994** - recognised brainstem death; **amended 2011** (Transplantation of Human Organs and Tissues Act); **THOTA Rules 2014**.
- Certified by a **Board of Medical Experts (4 members)**: (i) registered medical practitioner in charge of the hospital, (ii) an independent registered medical practitioner/specialist nominated from the approved panel, (iii) a neurologist or neurosurgeon from the panel (if unavailable - anaesthetist/intensivist/physician/surgeon nominated), (iv) the doctor treating the patient.
- **Two sets of tests, at least 6 hours apart** (Form 10); transplant team members must not be on the board; **mandatory request** to families of brainstem-dead patients in ICU.

Donor management (kidney-focussed)

- **"Rule of 100s"**: SBP >100 mmHg, urine output >100 mL/h, PaO₂ >100 mmHg, Hb >100 g/L.
- Volume (avoid over-hydration), vasopressors - **vasopressin** preferred (treats DI, spares catecholamines), noradrenaline; **desmopressin** for DI; hormone resuscitation (methylprednisolone, thyroid hormone, insulin) in unstable donors; correct Na⁺ (hypernatraemia linked to graft dysfunction in liver), warm to normothermia, lung-protective ventilation.
- Trials: low-dose **dopamine** donor pretreatment reduced dialysis need after transplant (Schnuelle, JAMA 2009); **mild donor hypothermia** reduced DGF (Niemann, NEJM 2015).

Diagram to draw: Herniation sequence with ICP curve; flowchart: storm → sympathetic loss → DI/hypothermia/DIC.

Recent advances

- Machine perfusion (hypothermic) of DBD/ECD kidneys reduces DGF; normothermic regional perfusion in DCD; Indian NOTTO national registry and "One Nation One Policy" allocation reforms.

Key points to remember: - Sequence: $\uparrow$ ICP $\rightarrow$ Cushing reflex $\rightarrow$ autonomic storm $\rightarrow$ loss of sympathetic tone (hypotension), DI, hypothermia, DIC. - Exclude drugs, hypothermia ($<35^{\circ}\text{C}$), metabolic causes before testing. - Apnoea test: $\text{PaCO}_2 \geq 60$ mmHg with no respiratory effort. - India: THOA 1994 (2011 amendment), Rules 2014 - 4-member board, two tests ≥ 6 h apart. - Rule of 100s; vasopressin for DI/hypotension.

I-3e. Male reproductive immunology

Type: SAQ (10 marks) | ~550 words

Introduction

- Spermatozoa appear at puberty – long after central immune tolerance is established – so meiotic/post-meiotic germ cell antigens are "**foreign**" ("autoantigenic"). Testis is an **immune-privileged site** (tolerates allografts). Campbell: Male reproductive physiology; Male infertility chapters.

Mechanisms of immune privilege

1. Blood-testis barrier (BTB)

- Formed by **tight junctions between adjacent Sertoli cells** (claudin-11, occludin, ZO-1) plus adherens and gap junctions; established at **puberty**.
- Divides seminiferous epithelium into **basal compartment** (spermatogonia, preleptotene spermatocytes – exposed to blood) and **adluminal compartment** (later spermatocytes, spermatids – sequestered).
- Dynamically opens/closes to let preleptotene spermatocytes migrate (stage VIII).
- Weak points:** rete testis and efferent ducts/epididymis – barrier less tight; common site of antigen leak.

2. Active immunoregulation (barrier alone is insufficient)

- Testicular macrophages** – predominantly M2-like, produce IL-10, low pro-inflammatory output.
- Regulatory T cells (Tregs)** and tolerogenic dendritic cells; peripheral tolerance to some meiotic antigens that are exposed.
- Sertoli cell** products – TGF- β , activin, protease inhibitors, complement inhibitors; **Leydig cell androgens** are immunosuppressive.
- Indoleamine 2,3-dioxygenase (IDO), galectin-1, PD-L1 expression.
- Fas ligand** on Sertoli cells inducing apoptosis of activated T cells – proposed but **debated/controversial**.
- Seminal plasma immunosuppressive factors (prostaglandin E, TGF- β) protect sperm in female tract.

Antisperm antibodies (ASA)

Causes (breach of BTB/excurrent duct)

- Vasectomy** (most common – ASA in majority of men afterward; relevant to vasectomy reversal outcomes), obstruction (CBAVD, epididymal), **testicular trauma, torsion**, testicular biopsy/surgery, **infection** (epididymo-orchitis, prostatitis, STI), cryptorchidism, possibly varicocele.
- IgG (serum + seminal) and IgA (local, more clinically significant in semen); also in female partner.

Effects

- Sperm **agglutination** (head-head, tail-tail), impaired motility and **cervical mucus penetration ("shaking phenomenon")**, impaired capacitation/acrosome reaction, **zona binding and fusion** failure, complement-mediated immobilisation.
- **Head-directed** antibodies more significant than tail-tip binding.

Testing (direct tests on sperm)

Test	Principle
Mixed antiglobulin reaction (MAR)	Fresh semen + IgG/IgA-coated latex beads/RBCs + antiglobulin → mixed agglutinates with motile sperm
Immunobead test (IBT)	Washed sperm + beads coated with anti-human IgG/IgA/IgM; reports % and location
Indirect	Detect ASA in serum/seminal plasma/cervical mucus using donor sperm

- **WHO 2010 manual: $\geq 50\%$ motile spermatozoa with bound particles** = clinically significant (immunological infertility); $< 50\%$ = reference limit.
- Indications: isolated asthenozoospermia, agglutination on semen analysis, unexplained infertility, post-trauma/reversal.

Treatment

Option	Comment
Corticosteroids (e.g. prednisolone)	Inconsistent benefit; risk of avascular necrosis of femoral head , GI bleed – generally not recommended
Sperm washing + IUI	Bypasses cervical mucus; modest benefit
IVF/ICSI	ICSI preferred – fertilisation largely unaffected by ASA (EAU male infertility)
Treat cause	Treat infection; microsurgical reconstruction

Inflammatory states

- **Leukocytospermia: $> 1 \times 10^6$ peroxidase-positive leukocytes/mL** (WHO) – reactive oxygen species → DNA fragmentation, lipid peroxidation of sperm membrane; evaluate for male accessory gland infection (semen culture, STI PCR); antibiotics if infection; antioxidant evidence limited.
- **Orchitis:**
 - Viral – **mumps** (post-pubertal; unilateral mostly; atrophy, subfertility), others incl. SARS-CoV-2 (orchitis/ACE2 expression – evolving data).
 - Bacterial epididymo-orchitis – *C. trachomatis*/*N. gonorrhoeae* (< 35 y), *E. coli* (older/urinary).
 - Granulomatous (TB, idiopathic).
- **Autoimmune orchitis: primary** (isolated, ASA + azoospermia, no systemic disease) vs **secondary** (systemic vasculitis – polyarteritis nodosa, Behçet, Henoch-Schönlein). Experimental autoimmune orchitis model (immunisation with testicular homogenate) shows lymphocytic infiltration and aspermatogenesis.
- Testicular cancer risk link: chronic inflammation in GCNIS-related lesions (research).

Diagram to draw: Cross-section of seminiferous tubule showing Sertoli tight junctions dividing basal/adluminal compartments, with Leydig cells/macrophages in interstitium.

Recent advances

- Sperm DNA fragmentation testing (TUNEL, SCSA) in men with leukocytospermia/ASA; testicular sperm for ICSI in high DFI.
- Immune checkpoint inhibitor-associated orchitis and hypogonadism; post-COVID orchitis data.

Key points to remember: - BTB = Sertoli-Sertoli tight junctions; basal vs adluminal compartments; formed at puberty. - Privilege also active: M2 macrophages, Tregs, TGF- β , androgens; FasL role debated. - ASA: vasectomy commonest; MAR/IBT $\geq 50\%$ motile sperm bound = significant (WHO). - Treatment: ICSI preferred; steroids out of favour (avascular necrosis). - Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks) | ~600 words

Classification (EAU Urological Infections 2024)

- **Uncomplicated** - acute, sporadic/recurrent cystitis or pyelonephritis in non-pregnant women without anatomical/functional abnormality or comorbidity.
- **Complicated** - men, pregnancy, anatomical/functional abnormality, catheter, renal disease, immunosuppression (diabetes, transplant).
- **Recurrent** - ≥ 2 in 6 months or ≥ 3 in 12 months.
- **Catheter-associated UTI (CA-UTI); urosepsis** (Sepsis-3: life-threatening organ dysfunction, qSOFA).

Principles

- **1. Confirm diagnosis, culture** (except simple cystitis in women).
- **2. PK/PD:**

Pattern	Drugs	Dosing goal
Time-dependent ($T > MIC$)	β -lactams, carbapenems	Frequent dosing/extended infusion
Concentration-dependent (C_{max}/MIC , AUC/MIC)	Aminoglycosides, fluoroquinolones	High once-daily dose; post-antibiotic effect

- **Urinary vs tissue levels:** nitrofurantoin/fosfomycin - high urinary, **low tissue levels** $\rightarrow$ **cystitis only**, never pyelonephritis/prostatitis/urosepsis. FQ, TMP-SMX penetrate prostate; β -lactams poorly.
- **Renal function:** adjust doses; avoid nitrofurantoin at $eGFR < 30$ mL/min.
- **3. Local resistance data** - use empirical agent only if local resistance **$< 20\%$** for cystitis (TMP-SMX, cephalosporins); FQ for pyelonephritis only if resistance **$< 10\%$** .
- **4. Collateral damage** - avoid FQs/cephalosporins for uncomplicated cystitis (ESBL, *C. difficile*; EMA FQ restrictions - tendinopathy, aortic aneurysm, neuropathy).
- **5. Shortest effective duration**, de-escalate to culture, IV $\rightarrow$ oral switch, **source control** (stent/PCN for obstructed infected kidney).
- **6. Pregnancy:** avoid FQ, tetracyclines; TMP in 1st trimester; nitrofurantoin near term.
- **7. Stewardship** - right drug, dose, route, duration; "start smart then focus"; don't treat ASB; local antibiograms.

Empirical choices (EAU 2024 / IDSA)

Scenario	First-line	Duration
Uncomplicated cystitis (women)	Fosfomycin trometamol 3 g single dose; nitrofurantoin (macrocrystal 50–100 mg qid, or monohydrate/macrocrystal 100 mg bd); pivmecillinam 400 mg tds	Fosfomycin 1 day; nitrofurantoin 5 d ; pivmecillinam 3–5 d
Alternatives (if local resistance <20%)	TMP-SMX 160/800 mg bd; trimethoprim 200 mg bd; cefadroxil 500 mg bd	TMP-SMX 3 d; TMP 5 d; cefadroxil 3 d
Cystitis in men	TMP-SMX or FQ per culture	7 d
Uncomplicated pyelonephritis (oral)	Ciprofloxacin 500–750 mg bd (7 d) or levofloxacin 750 mg od (5 d); alternatives: cefpodoxime 200 mg bd/ceftibuten 400 mg od (10 d), TMP-SMX (14 d)	5–14 d
Pyelonephritis (IV)	Ceftriaxone 1–2 g od, cefotaxime, ciprofloxacin/levofloxacin, gentamicin 5 mg/kg od, amikacin 15 mg/kg od; carbapenems/new BL-BLI only for proven/suspected MDR	Switch to oral when afebrile
Complicated UTI	Amoxicillin + aminoglycoside, or 2nd-gen cephalosporin + aminoglycoside, or IV 3rd-gen cephalosporin; FQ only if low resistance & not recently used	7–14 d (14 d in men when prostatitis not excluded)
CA-UTI	Change catheter if in >2 weeks before culture	7 d (prompt response), 10–14 d (delayed)
Urosepsis	Broad-spectrum IV within 1 h + source control	Per response

Asymptomatic bacteriuria (ASB)

- **Treat only in pregnancy** and **before urological procedures breaching the mucosa** (e.g. TURP/TURBT, ureteroscopy).
- **Do not treat:** healthy women, elderly, diabetics, catheterised, spinal cord injury, renal transplant (>2 months).

Prophylaxis

- **Recurrent UTI** (after behavioural advice and non-antimicrobial measures): vaginal oestrogen (postmenopausal), **immunoactive OM-89 (Uro-Vaxom)**, D-mannose, cranberry, ↑fluid intake, **methenamine hippurate** (ALTAR RCT, BMJ 2022 – non-inferior to antibiotic prophylaxis).
- Continuous low-dose (6–12 months): nitrofurantoin 50–100 mg od, fosfomycin 3 g every 10 days, trimethoprim 100 mg od, cephalexin 125–250 mg od; or **post-coital**; self-start therapy for compliant patients.
- **Peri-operative:** single dose/≤24 h – TURP, TURBT (large/high-risk), URS, PCNL; prefer **transperineal prostate biopsy** to reduce infection; no prophylaxis for uncomplicated cystoscopy/urodynamics.

Recent advances

- **Pivmecillinam** FDA-approved for uUTI (2024); **sulopenem etzadroxil/probenecid** (oral penem) FDA-approved for uUTI in women with limited options (2024); **gepotidacin** (first-in-class triazaacenaphthylene; EAGLE-2/3 trials) FDA-approved for uUTI (2025).
- cUTI agents: ceftazidime-avibactam, meropenem-vaborbactam, ceftolozane-tazobactam, **cefiderocol**, plazomicin, imipenem-relebactam; **cefepime-taniborbactam** (CERTAIN-1 trial in cUTI) – regulatory review ongoing.
- Vaccines (MV140 sublingual) and phage therapy under evaluation.

Diagram to draw: PK/PD curve showing C_{max}/MIC, AUC/MIC and T>MIC.

Key points to remember: - Recurrent UTI = $\geq 2/6$ months or $\geq 3/12$ months. - First-line cystitis: fosfomycin 3 g single, nitrofurantoin 5 d, pivmecillinam 3–5 d; TMP-SMX 3 d only if resistance $<20\%$. - Pyelonephritis: cipro 7 d / levo 750 mg 5 d (if FQ resistance $<10\%$); nitrofurantoin/fosfomycin never for pyelonephritis. - ASB: treat only in pregnancy and before mucosa-breaching urological surgery. - Aminoglycosides/FQ concentration-dependent; β -lactams time-dependent.